California Climate Analysis

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Math 586: Climate Mathematics

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1 Abstract

This study investigates the long-term climate trends in California over the past 75 years, focusing on temperature, precipitation, and greenhouse gas emissions. Using historical climate data from the National Oceanic and Atmospheric Administration (NOAA) and emissions data from the California Air Resources Board (CARB), the analysis centers on three coastal cities—San Diego, San Francisco, and Eureka—to represent Southern, Central, and Northern California respectively. Through Python-based data analysis and visualization, the research reveals a general warming trend across all regions, with San Francisco showing the most consistent increases in temperature and a concurrent decline in precipitation. While greenhouse gas emissions have decreased annually since 2000 due to policy interventions, the cumulative effect of long-lived gases such as CO_2 suggests an ongoing impact on climate. The findings highlight regional variations in climate response and suggest that rising temperatures in California are likely correlated with the accumulation of greenhouse gases, though the relationship with precipitation remains complex and region-specific. This work contributes to a clearer understanding of California's climate evolution and supports informed environmental policy and public awareness.

2 Introduction

The climate and weather are ongoing topics of discussion in the world today. The discussion often centers around climate change, and that opens the discussion towards the idea of greenhouse gases and their effect on the climate. The effect of greenhouse gases falls on the idea of periodic elements and molecules that are released into the ozone and that have a greater heat capacity than the elements that make up the majority of the world's ozone layer. Greenhouse gases such as carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O) are known to trap heat in the atmosphere by absorbing and re-emitting infrared radiation. Unlike the more abundant atmospheric gases such as nitrogen, oxygen, and argon, greenhouse gases interact more strongly with thermal radiation. Even small increases in their concentration can lead to significant changes in the Earth's energy balance, contributing to long-term warming trends.

This knowledge of the greenhouse gas effect has brought forward a question. Has California gotten hotter in the past 75 years, and is there correlation between the temperature, precipitation, and greenhouse gases?

To explore this issue, I employed a quantitative research method involving the analysis of long-term historical climate data. Temperature and precipitation records were obtained from the National Oceanic and Atmospheric Administration (NOAA), while green-

house gas emission data was sourced from the California Air Resources Board (CARB). By comparing trends in these datasets over a 75-year period, I aim to determine whether there is a statistically significant relationship between rising temperatures, changes in precipitation, and greenhouse gas levels.

Also employed is the use of python code to get the analysis and visual graphing.

This research is intended to provide clear, data-driven insight into the relationship between climate indicators in California, helping both the public and policymakers make informed decisions and communicate climate trends more effectively.

Ultimately, we had these questions in mind: Why is California so hot?

1. How much warmer has California been in the last 75 years?
2. Is California getting wetter or dryer in the last 75 years?
3. What is the trend of California's CO2 data?
4. Are there any correlations among the three?

3 Data and Method

3.1 Methodology

For this analysis we used 3 coastal California living centers to get a general basis of California's climate over 75 years in relatively similar areas. These 3 areas are, San Diego, San Francisco, and Eureka of

Humboldt county. These 3 cities would simulate coastal southern, central, and northern California.

With our three counties in mind we can now go about the data extraction. The two primary sources for the analyzed data sets were NOAA and CARB. NOAA providing 3 important forms of data, average daily temperature per year, average daily max temperature per year, and the total precipitation per year. CARB provided the yearly Million Metric Tons of Carbon Dioxide Equivalent (MMTCO2e) from 2000 through 2023 which were the only years currently available for accurate data of greenhouse gas emissions.

The use of each dataset was thought about with questions in mind. Average daily temperature is a simple direct answer to our problem, is California getting hotter? In correlation, we took the average daily max temperature to take into account the feeling that the hottest days of the year may be influencing our perception of weather. To analyze both the average daily temperature, and the average daily max temperature can also provide insight into the dissipation of heat. Whether the day as a whole is hotter, through the night, or that just the period with sunlight such a high influence on the perception of weather. We are looking to get an understanding of whether nights are keeping more heat from the sunlight periods, or if there is a stark difference in nighttime and daytime temperatures.

3.2 Data and Graphs

The first set of data points to get into is the the Average Daily Temperature over a 75 year period. For this data we took the NOAA datasets of San Diego, San Francisco, and Eureka. Then isolated the wanted dataset variables with code like:

$$\text{SanDiego} = \text{datas} + \text{"SDYearly43to25.csv"}$$

$$\text{SD} = \text{pd.readcsv(SanDiego)}$$

$$\text{SDtavg} = \text{SD.iloc[12:86,60].values}$$

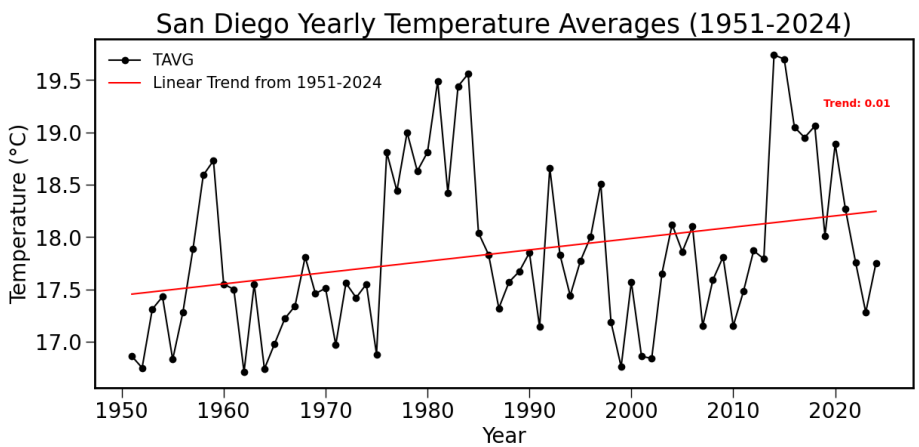

Figure 1: In this graph we can see a general increase in San Diego's average temperature over the last 75 years, with a linear trend of 0.01 degrees Celsius per year, or 0.75 over the 75 year period.

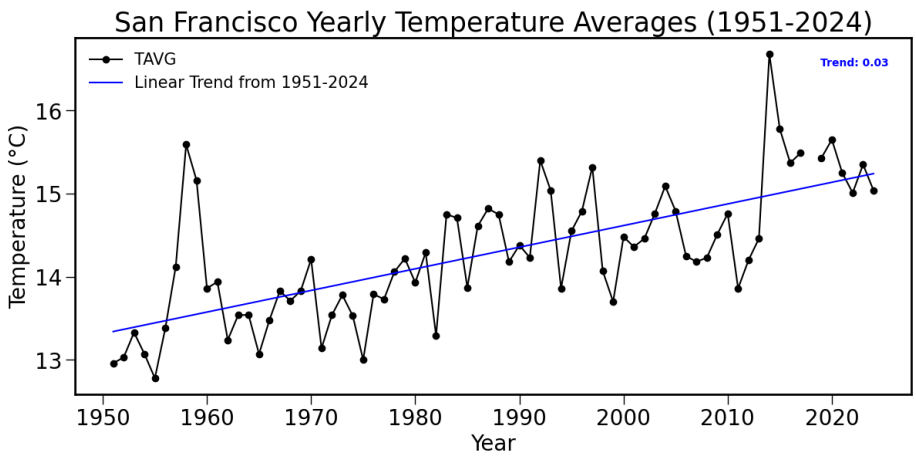

Figure 2: In this graph we can see a general increase in San Francisco's average temperature over the last 75 years, with a linear trend of 0.03 degrees Celsius per year, or 0.75 over the 75 year period.

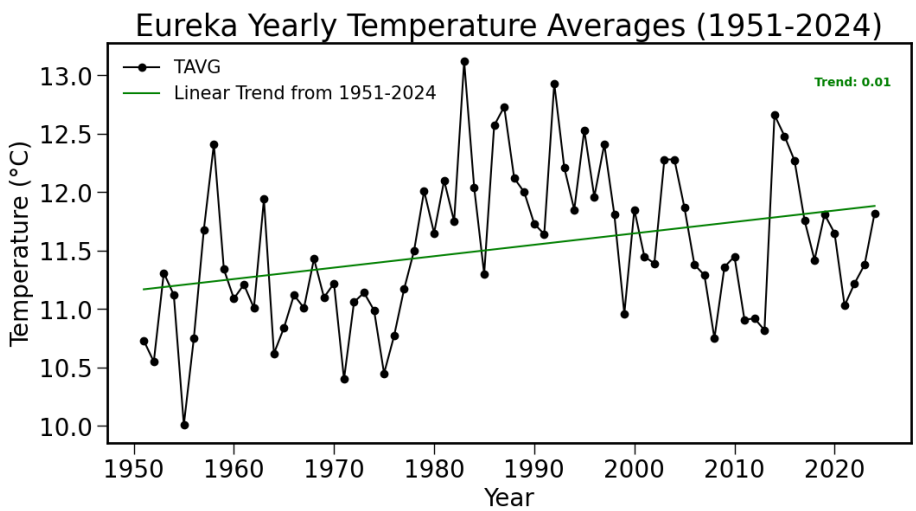

Figure 3: In this graph we can see a general increase in Eureka CA's average temperature over the last 75 years, with a linear trend of 0.01 degrees Celsius per year, or 0.75 over the 75 year period.

Now with the average daily temperature done and visually presented we moved toward the daily max temperature over a 75 year period. This code would be pivoted from the daily average like so:

SDtmax = SD.iloc[12:86,62].values

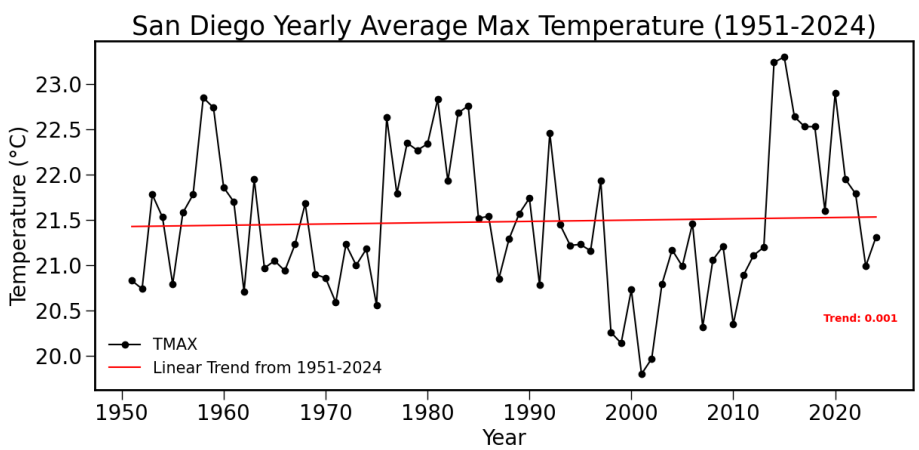

Figure 4: In this graph we can see a slight increase in San Diego's average max temperature over the last 75 years, with a linear trend of 0.001 per year, or 0.75 degrees Celsius over the 75 year period.

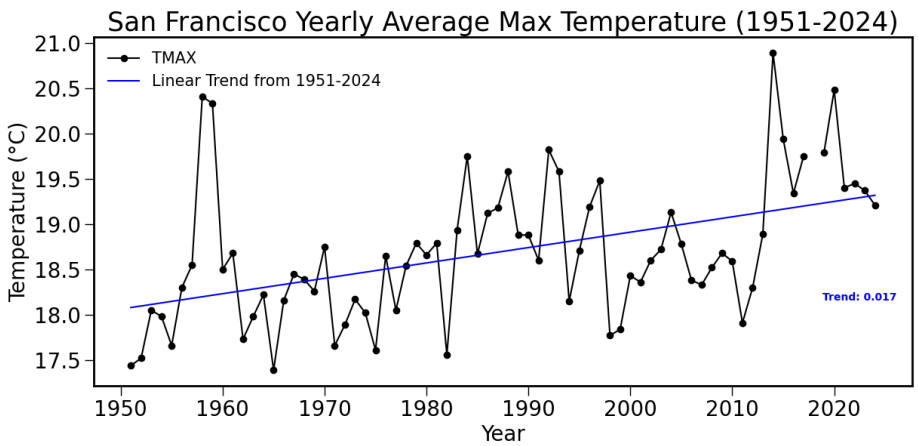

Figure 5: In this graph we can see a more notable increase in San Francisco's average max temperature over the last 75 years, with a linear trend of 0.017 degrees Celsius per year, or 0.75 over the 75 year period.

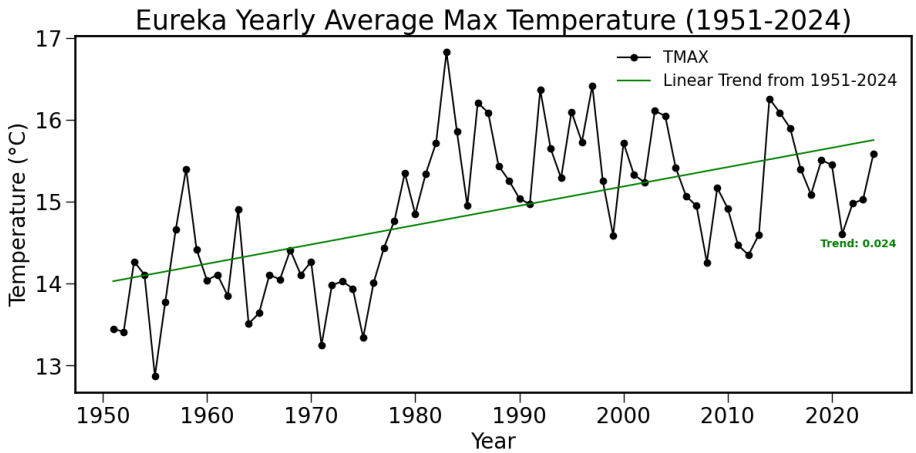

Figure 6: In this graph we can see an even more substantial increase in Eureka CA.'s max temperature over the last 75 years, with a linear trend of 0.024 degrees Celsius per year, or 0.75 over the 75 year period.

The last major dataset variable from the NOAA data model of importance to us was the total precipitation over the same 75 year period. Note that the metric of measurement has changed from the teens to the hundreds so although the trend appears to be larger it

is not relatively.

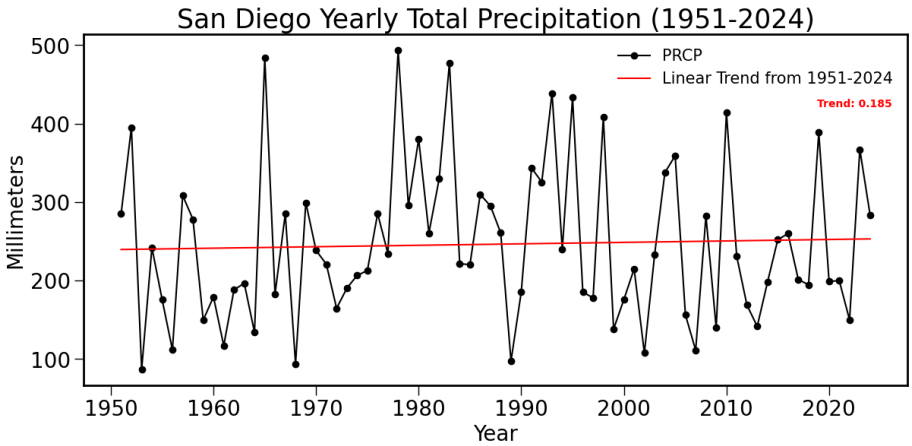

Figure 7: In this graph we can see a slight increase in total precipitation in San Diego over the last 75 years, with a linear trend of 0.185 mm of precipitation per year.

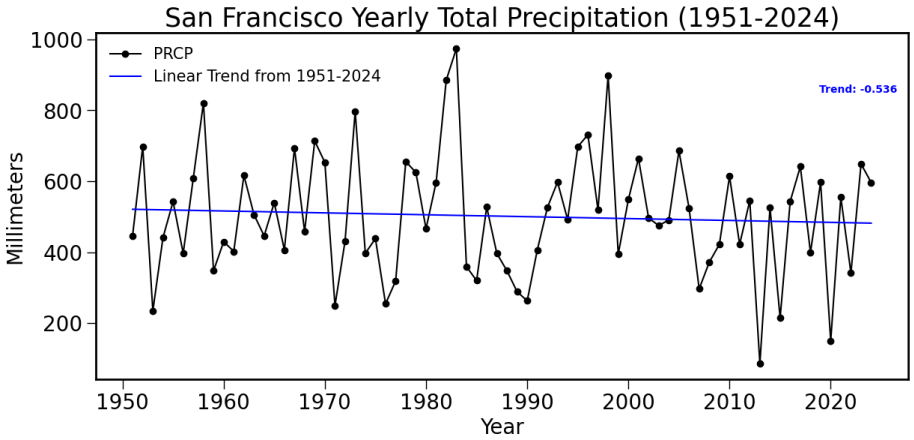

Figure 8: In this graph we can see a notable decrease in total precipitation in San Francisco over the last 75 years, with a linear trend of -0.536 mm of precipitation per year, or 0.75 over the 75 year period.

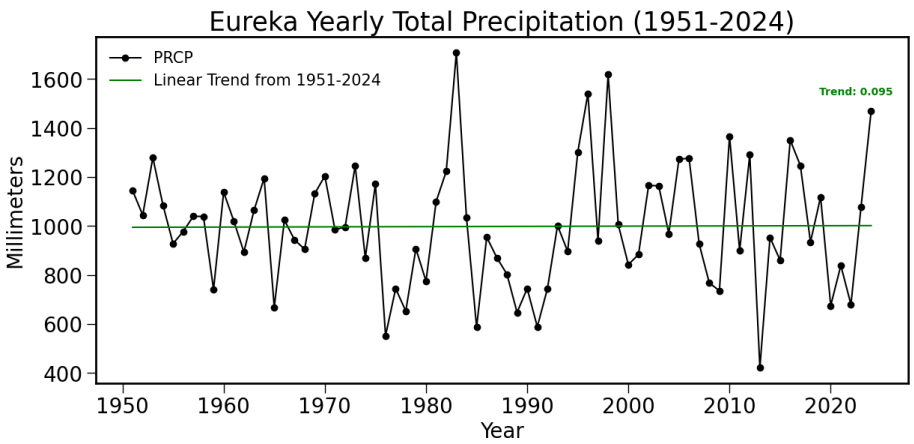

Figure 9: In this graph we can see a very slight increase in total precipitation in Eureka over the last 75 years, with a linear trend of 0.095 mm of precipitation per year.

Lastly, we use the CARB data over MMTCO2e, to get a visual understanding of the carbon emission rate in California. Note that the dataset refers to yearly carbon equivalent emissions, and not total greenhouse gas weight. Also note, that the period of study is 2000-2023.

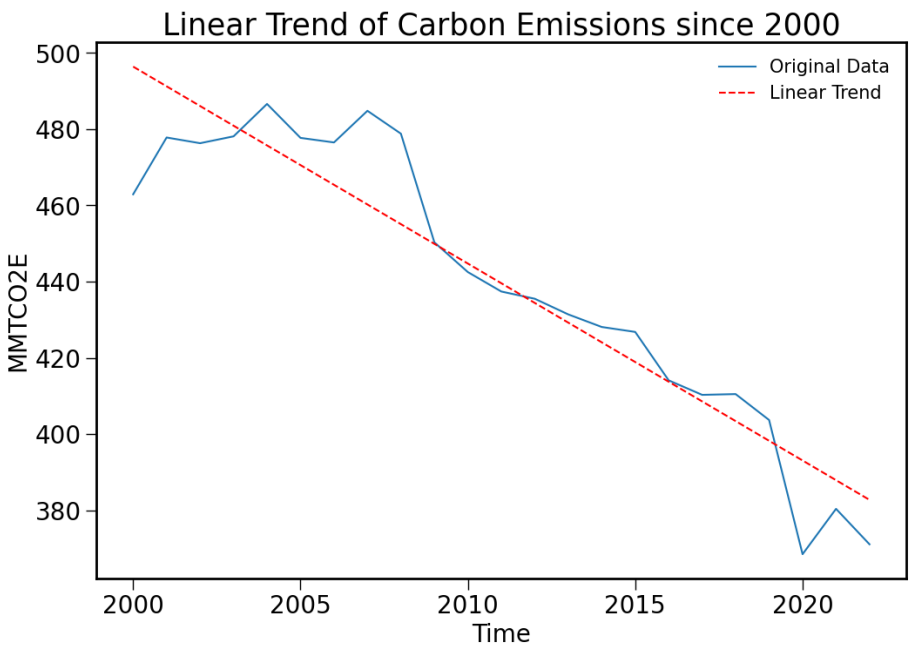

Figure 10: In this graph we can see a strong trend downward, as State legislation have enforced cleaner air policies.

4 Results

Returning to our Primary and 4 Sub-questions:

Why is California so hot?

1. How much warmer has California been in the last 75 years?
2. Is California getting wetter or dryer in the last 75 years?
3. What is the trend of California's CO2 data?
4. Are there any correlations among the three?

We can now answer, that California has been getting warmer over the last 75 years. Although it is not an equal change throughout the state. Southern California due its closer proximity to the equator

is having a steady change, but due to more erratic climate patterns over the last 75 years the change is not as noticeable. San Diego our stand-in for Southern California has seen as low as a 3 degree celsius shift in max average and daily average temperature. So these high-low shifts make the trend of the temperature appear with less change. With the information that daily max temperatures can be upwards of 4 degrees higher than the daily average, this would imply a stark shift in temperature between night and day. Similarly precipitation totals have high contrasts from year to year which makes the trend appear low. Although notably San Diego has the lowest precipitation rate between itself, San Francisco, and Eureka being doubled and tripled respectively.

With our Central California stand-in San Francisco, we can see a noticeable correlation between all of our three primary variables of daily average temperature, average max temperature, and precipitation totals. With strong positive trends towards increasing daily and max temperatures, and a notable decline in precipitation totals.

Finally, our Northern California stand-in Eureka, is similar to San Francisco in the temperature trends, but still has a positive precipitation trend.

Now we take into account the trend of greenhouse gas emissions. Although the trend is negative, it is important to note that the data is referring to yearly emissions. With the rate that greenhouse gasses maintain stability for thousands of years, and the higher heat

retention of them. We can envision a total greenhouse gas trend that is only increasing. The million metric tons of Carbon Dioxide equivalent only compounds on top of itself.

With this we now have a larger total vision of California's rising warmth. Although there is legislative pressure to reduce greenhouse gasses, the continued release of the same gases has correlated with rising temperatures in three major California areas. This can be most prevalently seen in San Francisco that has had a steady temperature increase over the last 75 years.

There is no strong correlation with rising temperatures and precipitation. San Francisco shows a strong inverse relationship, however San Diego and Eureka show a near parallel relationship. In the case of San Francisco it appears that the dryness may exacerbate the heat, but not necessarily cause the rising temperature.

A correlation between precipitation and greenhouse gas emissions is a more unsure relationship. For Eureka and San Diego the total increase of precipitation relates with the continued emissions of greenhouse gas even to lesser extent by each year. For San Francisco there is a near direct correlation, between reduced precipitation and higher Carbon emission totals. We must take into account the global effect of Greenhouse Gasses with the melting of the polar ice caps. With San Francisco the more central location, perhaps the effect is less influenced.

5 Conclusion

This research set out to investigate whether California has experienced significant warming over the past 75 years, and if there are discernible correlations between temperature trends, precipitation changes, and greenhouse gas emissions. Using historical climate data from NOAA and greenhouse gas emissions data from CARB, we analyzed three representative coastal cities—San Diego, San Francisco, and Eureka—spanning southern, central, and northern California.

The analysis reveals a clear warming trend across all three regions, although the magnitude and patterns of change vary. San Francisco, in particular, exhibited the most consistent increases in both average and maximum temperatures alongside a notable decline in annual precipitation. In contrast, San Diego and Eureka showed more modest temperature increases and fluctuating precipitation trends, making clear conclusions more difficult.

Greenhouse gas emission data, while limited to the period from 2000 to 2023, show a declining trend in annual emissions—largely due to legislative efforts. However, given the long atmospheric lifetimes of gases like CO_2, the cumulative greenhouse gas burden continues to rise, likely contributing to the long-term warming observed.

Alternatives and Methodological Considerations

Alternative approaches could include the use of climate models

or satellite-based datasets to analyze temperature and greenhouse gas concentrations more holistically and at higher spatial resolution. Additionally, including inland cities or comparing urban versus rural data could shed light on the impact of urban heat islands or localized climate phenomena.

One could also incorporate atmospheric circulation models or consider global datasets to contextualize California's trends within a broader climate system.

Pros and Cons of This Method

Pros:

Data-driven approach: The use of long-term historical data provides an empirical foundation for identifying climate trends.

Geographic representation: Selecting three cities along California's coast offers insight into regional variation across the state.

Transparency and reproducibility: The use of Python code and publicly available datasets allows for replicability and further refinement.

Cons:

Limited greenhouse gas data timeframe: CARB emissions data only covers 23 years, restricting long-term correlation analyses.

Spatial limitations: Only coastal regions were analyzed; results may not generalize to inland California.

Simplified correlation assumptions: Correlation does not imply causation, and more sophisticated statistical or causal models could

provide deeper insight.

Final Thoughts:

Overall, the evidence suggests that California has warmed over the past 75 years, with greenhouse gas accumulation likely playing a significant role. While precipitation trends are less consistent and show weaker correlations with temperature, they still reflect potential regional climate shifts. This work underscores the importance of continued climate monitoring and suggests that even in the face of emission reductions, legacy effects from past emissions will persist in shaping the state's climate future.

6 Data and Code Availability

CODE and DATASETS used can be found at the following githubs. The Githubs are the same, just one with the required name.:

https://github.com/EvanNorme/Math586ResearchReport

https://github.com/EvanNorme/Climate-Math-Research

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The data sets for TAVG, TMAX, and Precipitation were taken from the NOAA dataset finder for San Diego, San Francisco, and Eureka California.

https://www.ncei.noaa.gov/access/search/data-search/global-summary-of-the-year

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Greenhouse gas emissions data was found through:

California Air Resources Board, "Current California GHG Emission Inventory Data 2000–2022," 2024. [Online].

Available: https://ww2.arb.ca.gov/ghg-inventory-data

7 References

Climate Data and Government Sources

- California Air Resources Board. (2024). *California greenhouse gas emission inventory.* https://ww2.arb.ca.gov/ghg-inventory-data
- National Centers for Environmental Information. (2024). *Global summary of the year dataset.* National Oceanic and Atmospheric Administration. https://www.ncei.noaa.gov/access/search/data-search/global-summary-of-the-year
- National Integrated Drought Information System. (2024). *California drought information.* https://www.drought.gov/states/california
- PRISM Climate Group. (2024). *PRISM climate data.* Oregon State University. https://prism.oregonstate.edu
- U.S. Environmental Protection Agency. (2023). *Inventory of U.S. greenhouse gas emissions and sinks.*

  https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks
- California Legislative Analyst's Office. (2018). *Climate change and Cap-and-Trade.* https://lao.ca.gov/reports/2018/3933/climate-policies-overview-020718.pdf

Peer-Reviewed Scientific Literature

- Diffenbaugh, N. S., Swain, D. L., Touma, D. (2015). Anthropogenic warming has increased drought risk in California. *Proceedings of the National Academy of Sciences, 112*(13), 3931–3936. https://doi.org/10.1073/pnas.1422385112

- Jones, C., Stott, P. A., Christidis, N. (2008). Human contribution to rapidly increasing summer temperatures in California. *Nature, 451*(7174), E1–E3. https://doi.org/10.1038/nature06559

- Swain, D. L. (2021). A tale of two Californias: The links between megadrought, climate change, and water security. *Nature Climate Change, 11*(7), 579–584. https://doi.org/10.1038/s41558-021-01004-8

- Vose, R. S., Easterling, D. R., Kunkel, K. E., et al. (2017). Temperature changes in the United States. In D. J. Wuebbles et al. (Eds.), *Climate Science Special Report: Fourth National Climate Assessment, Vol. 1* (pp. 185–206). U.S. Global Change Research Program. https://science2017.globalchange.gov

- Bedsworth, L. W., Hanak, E. (2013). *Preparing California for a changing climate*. Public Policy Institute of California.

  https://www.ppic.org/publication/preparing-california-for-a-changing-climate/

Statistical Methods and Tools

- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830. https://scikit-learn.org/stable/

- Seabold, S., Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with Python. In *Proceedings of the 9th Python in Science Conference* (pp. 92–96).

  https://conference.scipy.org/proceedings/scipy2010/pdfs/seabold.pdf

8 Peer Review

Peer Review For: Evan Michel By: Isabella Tabrizi

Your abstract does a great job stating the purpose and methods as well as conclusions made from the results. Overall it is a good executive summary. The intro explains well why it is a relevant topic to be discussing. It also gives necessary context about the science behind how greenhouse gases relate to climate change and global warming. I like that you had specific research questions listed. I appreciate how you detailed why the three cities were chosen. Next, I appreciate the insight of why it is important to look at maximum temperatures as well as averages. Your methodology does a fantastic job of explaining each decision that was made in your research. The data and graphs section analyzes the graphs well.

One suggestion I would make about your formatting is to make sure things are not spaced awkwardly on the page. For example, If a section has the header on one page and the body on the next, it might be a good idea to move the whole section to start on the next page.

Additionally, there seems to be a typo on Figure 8 in section 3.2. The graph is labeled for San Francisco but your caption says San Diego. Same with figure 9. I would also go into more detail about what is meant by "with a linear trend of 0.095 per year" for example. You could include some units for more context.

Lastly, I would suggest maybe explaining the code in more depth, such as explaining specific lines or functions briefly rather than just stating your goal and then showing the code. You could also show the code used to calculate the linear trends from the given data.